

Docker_Task:

- Use this Java application shared via this link:

<https://gitlab.com/seymuralye/myapp>

git clone <https://gitlab.com/seymuralye/myapp>

```
[root@rhel9 ~]# git clone https://gitlab.com/seymuralye/myapp
Cloning into 'myapp'...
warning: redirecting to https://gitlab.com/seymuralye/myapp.git/
remote: Enumerating objects: 38, done.
remote: Counting objects: 100% (38/38), done.
remote: Compressing objects: 100% (25/25), done.
remote: Total 38 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (38/38), 67.22 KiB | 33.61 MiB/s, done.
Resolving deltas: 100% (1/1), done.
[root@rhel9 ~]# ll
total 4
-rw-----. 1 root root 978 Jul 13 14:49 anaconda-ks.cfg
drwxr-xr-x. 4 root root  47 Jul 21 14:29 myapp
[root@rhel9 ~]#
```

tree myapp

```
[root@rhel9 ~]# tree myapp/
myapp/
├── demo
│   ├── mvnw
│   ├── mvnw.cmd
│   ├── pom.xml
│   └── README.md
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── com
│   │   │   │   ├── company
│   │   │   │   │   └── Demo
│   │   │   │   │       ├── config
│   │   │   │   │       │   ├── ApplicationProperties.java
│   │   │   │   │       │   ├── controller
│   │   │   │   │       │   │   ├── DateController.java
│   │   │   │   │       │   │   ├── DemoApplication.java
│   │   │   │   │       │   │   ├── model
│   │   │   │   │       │   │   │   ├── DateDto.java
│   │   │   │   │       │   │   │   └── service
│   │   │   │   │       │   │       └── DateService.java
│   │   └── resources
│   │       └── application.yml
│   └── test
│       ├── java
│       │   ├── com
│       │   │   ├── company
│       │   │   │   └── Demo
│       │   │   │       └── DemoApplicationTests.java
└── README.md

17 directories, 12 files
[root@rhel9 ~]#
```

- Create a **Dockerfile** that builds the code using the **Maven latest package** command. In the last line of the Dockerfile, use **CMD** to run the **JAR** file.

vim Dockerfile

```
# === Stage 1: Build the application ===  
FROM maven:3.8.8-eclipse-temurin-11 AS build  
# Set the working directory inside the container  
WORKDIR /app  
# Copy the necessary files for the build process  
COPY myapp/demo/pom.xml .  
COPY myapp/demo/src ./src  
# Build the project without running tests  
RUN mvn clean package -DskipTests  
  
# === Stage 2: Run the application ===  
FROM eclipse-temurin:11-jre-jammy  
# Set the working directory  
WORKDIR /app  
# Copy the compiled JAR file from the previous stage  
COPY --from=build /app/target/*.jar ./app.jar  
# Expose port (change if your app uses a different one)  
EXPOSE 8080  
# Run the application  
ENTRYPOINT ["java", "-jar", "app.jar"]
```

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# Expose port (change if your app uses a different one)  
EXPOSE 8080  
# Run the application  
ENTRYPOINT ["java", "-jar", "app.jar"]
```

docker build -t myapp:1.0 .

```
[root@rhel9 ~]# docker build -t myapp:1.0 .
[*] Building 200.6s (14/14) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 769B
=> [internal] load metadata for docker.io/library/eclipse-temurin:11-jre-jammy
=> [internal] load metadata for docker.io/library/maven:3.8.8-eclipse-temurin-11
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [build 1/5] FROM docker.io/library/maven:3.8.8-eclipse-temurin-11@sha256:a9023829d06570e9e4f99928b360aca6a6ebec642901fa15b69726221dd2eda4
=> resolve docker.io/library/maven:3.8.8-eclipse-temurin-11@sha256:a9023829d06570e9e4f99928b360aca6a6ebec642901fa15b69726221dd2eda4
=> sha256:1b06fec4774482ac82ddb5d8a91c94986949961f6dbb2069c7a7aa087655c226 156B / 156B
=> sha256:ff547c18fe9143e96d55a15fb9cc57e6ec52f6db36a65a4a19475fd3237babab 356B / 356B
=> sha256:4852df4714fe2f033ee6410c9e1022d53bffcab61a3b6135830db35c590b787d 850B / 850B
=> sha256:750b98e3a39f992e416a4ab683462918af31f3091cc7a30758e6bc645d24330b 8.36MB / 8.36MB
=> sha256:c06a84945f550496044b8478391d9399846ab31f5b14b2d381254a1aa08f8e3e 734B / 734B
=> sha256:3240050f08dc26d068227d57cf1c291264d7e2ff3820e3bf694aacdc3038036 18.99MB / 18.99MB
=> sha256:1bcc0786dd85cc196d864618ed34d937a23b183fcd6b6a1610fe3b97a2fbdff 174B / 174B
=> sha256:fcd82cbe4eb323290897e9d752b7b4f95a52dd4c4f2affce239a93b1fda268d 145.27MB / 145.27MB
=> sha256:3dd181f9be599de628e1bcbdb68d517125e07f968824bcf7b7ed8d28ad1026b1 30.45MB / 30.45MB
=> sha256:6d7336e2190966050b2455fbecac42788c07045994fb8ce47da65dbdc145020e 12.90MB / 12.90MB
=> extracting sha256:3dd181f9be599de628e1bcbdb68d517125e07f968824bcf7b7ed8d28ad1026b1
=> extracting sha256:6d7336e2190966050b2455fbecac42788c07045994fb8ce47da65dbdc145020e
=> extracting sha256:fcd82cbe4eb323290897e9d752b7b4f95a52dd4c4f2affce239a93b1fda268d
=> extracting sha256:1bcc0786dd85cc196d864618ed34d937a23b183fcd6b6a1610fe3b97a2fbdff
=> extracting sha256:c06a84945f550496044b8478391d9399846ab31f5b14b2d381254a1aa08f8e3e
=> extracting sha256:3240050f08dc26d068227d57cf1c291264d7e2ff3820e3bf694aacdc3038036
=> extracting sha256:750b98e3a39f992e416a4ab683462918af31f3091cc7a30758e6bc645d24330b
=> extracting sha256:4852df4714fe2f033ee6410c9e1022d53bffcab61a3b6135830db35c590b787d
=> extracting sha256:ff547c18fe9143e96d55a15fb9cc57e6ec52f6db36a65a4a19475fd3237babab
=> extracting sha256:1b06fec4774482ac82ddb5d8a91c94986949961f6dbb2069c7a7aa087655c226
=> [stage-1 1/3] FROM docker.io/library/eclipse-temurin:11-jre-jammy@sha256:76ffb747ad3a62a8b81ac3c76e2c3b1c06e475b1e10109d0a4dd604db627c9f5
=> resolve docker.io/library/eclipse-temurin:11-jre-jammy@sha256:76ffb747ad3a62a8b81ac3c76e2c3b1c06e475b1e10109d0a4dd604db627c9f5
=> sha256:4e00f30580e97c4a9f3935a1c91e783e43210534924feb464df388610d355569 2.28kB / 2.28kB
=> sha256:655078516f617ac3d2abbc3df76cfd0cba56371452d0fbb68c34b53d372cf87 159B / 159B
=> sha256:e0cbf5e8f8832bc6005302e9c37ffcf2f813201c42f3ac565971a06b98ec7509e 47.34MB / 47.34MB
=> sha256:c480354d9994f146cc3907f464a2bcbf5f8c881af89bc39c539790868da0d948 16.11MB / 16.11MB
=> sha256:d6834b4a794c03efa2c998853e64969fa8851b11b2ade63292268872a37759d0 29.74MB / 29.74MB
=> extracting sha256:d6834b4a794c03efa2c998853e64969fa8851b11b2ade63292268872a37759d0
=> extracting sha256:c480354d9994f146cc3907f464a2bcbf5f8c881af89bc39c539790868da0d948
=> extracting sha256:e0cbf5e8f8832bc6005302e9c37ffcf2f813201c42f3ac565971a06b98ec7509e
=> extracting sha256:655078516f617ac3d2abbc3df76cfd0cba56371452d0fbb68c34b53d372cf87
=> extracting sha256:4e00f30580e97c4a9f3935a1c91e783e43210534924feb464df388610d355569
=> [internal] load build context
=> => transferring context: 7.24kB
=> [stage-1 2/3] WORKDIR /app
=> [build 2/5] WORKDIR /app
=> [build 3/5] COPY myapp/demo/pom.xml .
=> [build 4/5] COPY myapp/demo/src ./src
=> [build 5/5] RUN mvn clean package -DskipTests
=> [stage-1 3/3] COPY --from=build /app/target/*.jar ./app.jar
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:6efeb5c49d6b88f79b333d418c3e31284f875e49c40819364025889a33ff615
=> => exporting config sha256:0610f4af5a74a088a7974b11a068b13faad4c0e9cf3a896e1915a711c3ce268f
=> => exporting attestation manifest sha256:c2beaa30612b48c8ea1fa9b9beb458c1bdf0fce604c3d6020487d111afeedd195
=> => exporting manifest list sha256:703452b779b2b5c1c93f6813b3ab2ea60fcf839b56878ad795801900011830c6
=> => naming to docker.io/library/myapp:1.0
=> => unpacking to docker.io/library/myapp:1.0
[root@rhel9 ~]#
```

docker images

```
[root@rhel9 myapp]# docker images
INFO In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
myapp:1.0 c95aa967f4b9 415MB 115MB
```

- Run this image.

docker run -d --name myapp -p 8080:8080 myapp:1.0

docker ps

```
[root@rhel9 myapp]# docker run -d --name myapp -p 8080:8080 myapp:1.0
d618519d52e770303077e4f3d9afb353a5d2988e74bf34800155498d742162c
[root@rhel9 myapp]# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
d618519d52e7 myapp:1.0 "java -jar app.jar" 7 seconds ago Up 7 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp myapp
```

- If everything is correct, you can verify it with `curl IP:PORT/api/v1/date` (don't forget to take a screenshot of your result).`

curl 172.17.0.2:8080/api/v1/date

```
[root@rhel9 myapp]# curl 172.17.0.2:8080/api/v1/date
{"env":"demo-name","date":"2026-07-21T10:58:42.021932"}[root@rhel9 myapp]#
```

